
Integrable Systems — Exercise Sheet 17

Dressing and the thermodynamic Bethe ansatz

Variant: Questions only

Conventions, as in Lecture 17, for the gas ($\hbar = 2m = 1$, $c > 0$): $K(k) = 2c/(k^2 + c^2)$, bare momentum $p_0 = k$ and bare energy $\varepsilon_0 = k^2$; the filling fraction $\vartheta = \rho/\rho_t$, $\rho_t = \rho + \rho_h$, and the dressing $f^{\text{dr}} = f + \frac{1}{2\pi} \int K(\cdot - k') \vartheta(k') f^{\text{dr}}(k') dk'$; the density equation $\rho_t = \frac{1}{2\pi} + \frac{1}{2\pi} \int K(\cdot - k') \rho(k') dk'$. At $T = 0$ the sea is $[-q, q]$, ϑ its indicator, and the dressed energy solves $\varepsilon(k) - \frac{1}{2\pi} \int_{-q}^q K(k - k') \varepsilon(k') dk' = k^2 - \mu$ with $\varepsilon(\pm q) = 0$. At $T > 0$ the Yang–Yang equation is $\varepsilon = k^2 - \mu - \frac{T}{2\pi} \int K(k - k') \ln(1 + e^{-\varepsilon(k')/T}) dk'$, with $\vartheta = 1/(1 + e^{\varepsilon/T})$ and pressure $P = \frac{T}{2\pi} \int \ln(1 + e^{-\varepsilon/T}) dk$; P_F is the free-fermion pressure, the same with $\varepsilon = k^2 - \mu$.

Exercise 1 (dressing). (i) Show that $2\pi\rho_t = (p'_0)^{\text{dr}}$ for every ϑ . (ii) At $T = 0$, with $\varepsilon(\pm q) = 0$, show that $\varepsilon' = (\varepsilon'_0)^{\text{dr}}$ on $(-q, q)$. (iii) The edge condition matters: for $f(k) = k$, which does not vanish at $\pm q$, compute $(f^{\text{dr}})'(0)$ and $(f')^{\text{dr}}(0)$ at $c = q = 1$.

Exercise 2 (counting). (i) Show that $\ln \binom{A}{B} = A \ln A - B \ln B - (A - B) \ln(A - B) + O(\ln A)$ for $0 < B < A$. (ii) Deduce Lecture 17's entropy density $s = \int [\rho_t \ln \rho_t - \rho \ln \rho - \rho_h \ln \rho_h]$, and show that its integrand is $\rho_t \sigma(\vartheta)$ with $\sigma(\vartheta) = -\vartheta \ln \vartheta - (1 - \vartheta) \ln(1 - \vartheta)$. (iii) Where is σ largest, and what is it at $\vartheta = 0$ and 1 ? (iv) Show that $F(x, y) = y \sigma(x/y)$ is concave on $0 < x < y$, and deduce that s is concave in ρ (Lecture 17 uses this to show that the Yang–Yang point is a minimum).

Exercise 3 (the Yang–Yang equation: three limits). (i) With $K \rightarrow 0$ (impenetrable), show that the Yang–Yang equation gives free fermions: $\varepsilon = k^2 - \mu$, $\rho_t = \frac{1}{2\pi}$, and $n = \partial P_F / \partial \mu$ is the Fermi–Dirac density. (ii) As $\mu/T \rightarrow -\infty$ (dilute), show that $P_F/T^{3/2} \rightarrow e^{\mu/T}/(2\sqrt{\pi})$, Lecture 17's Boltzmann asymptote. (iii) With $k = \sqrt{T} \kappa$ and $c = \sqrt{T} \hat{c}$, show that ε/T depends on $(\kappa, \mu/T, \hat{c})$ only, and $P/T^{3/2}$ on $(\mu/T, \hat{c})$ only.

Exercise 4 (large c : a longer box). Lecture 17 proves that as $c \rightarrow \infty$, $\varepsilon(k) = k^2 - \mu - 2P/c + O(c^{-2})$ uniformly on compact sets. (i) Deduce that $P = P_F(\mu + 2P/c) + O(c^{-2})$, and compare with hard rods of length a , $P = P_F(\mu - aP)$: what is a ? (ii) (Computational.) At $c = 100$, $T = \mu = 1$, solve the Yang–Yang equation by the tutorial's iteration. By how much does P exceed $P_F(\mu)$, and how well does $P = P_F(\mu + 2P/c)$, solved for P , account for it?

Exercise 5 (Lieb's two branches). (Computational.) Take $\gamma = 4$ and $n = 1$ (so $c = 4$), at $T = 0$. (i) Find q (from $n = \int_{-q}^q \rho = 1$), μ (from $\varepsilon(\pm q) = 0$), and the sound velocity $v_s = \varepsilon'(q)/(2\pi\rho_t(q))$. (ii) With $p(k) = k + \int_{-q}^q \theta(k - k') \rho(k') dk'$, $\theta(k) = 2 \arctan(k/c)$, plot Type I, $(p(k) - p(q), \varepsilon(k))$ for $k > q$, and Type II, $(p(q) - p(k), -\varepsilon(k))$ for $|k| < q$. Show that Type II spans $0 \leq p \leq 2\pi n$, vanishing at both ends, and find its top. (iii) Show that both branches leave $p = 0$ with slope v_s .

Exercise 6 (solving the Yang–Yang equation). (Computational.) Take $c = 1$ and the μ of the $T = 0$ sea with $q = 1.5$. (i) At $T = 0$ give μ , γ , e and $v = \varepsilon'(q)/(2\pi\rho_t(q))$, and show that $P(0) = -\frac{1}{2\pi} \int_{-q}^q \varepsilon = \mu n - E/L$. (ii) At $T = 0.02$, solve the Yang–Yang equation at the same μ and compare $P(T) - P(0)$ with the Sommerfeld form $\pi T^2/(6v)$ of a Luttinger liquid with

central charge 1. (iii) At $T = 0.25$ check the thermodynamics: $n = \int \rho$ against $\partial P / \partial \mu$, and Lecture 17's s against $\partial P / \partial T$.

Exercise 7 ★. *Free spins in Takahashi's equations.* Lecture 17's reading gives, for the Heisenberg chain at $T > 0$ in a field $h \geq 0$, $\ln \eta_n = -\delta_{n1} \frac{\pi J}{T} s + s * \ln[(1 + \eta_{n-1})(1 + \eta_{n+1})]$, $\eta_0 \equiv 0$, $f = -J \ln 2 - T \int s \ln(1 + \eta_1)$, with $s(\lambda) = 1/(2 \cosh \pi \lambda)$ and the field entering only through $\lim_n (\ln \eta_n)/n = h/T$. (i) At $J = 0$ and $h = 0$, look for constant η_n . Show that they must satisfy $\eta_n^2 = (1 + \eta_{n-1})(1 + \eta_{n+1})$, and that $\eta_n = n(n+2)$ does, with $f = -T \ln 2$. (ii) Show that $1 + \eta_n = (\sinh((n+1)x)/\sinh x)^2$ also does, for any $x > 0$, with $(\ln \eta_n)/n \rightarrow 2x$ and $f = -T \ln(2 \cosh x)$. (iii) Compare with a free spin in $H = -h \sum_n S_n^z$, $h > 0$, and deduce why the boundary condition is h/T , not the $2h/T$ of a book whose field term is $-2h \sum_n S_n^z$. (iv) Show that, as long as its terms are positive, a solution of the recursion with $\eta_0 = 0$ is fixed by η_1 , and that $\eta_n > 0$ for every n only if $\eta_1 \geq 3$. Deduce that the positive constant solutions are exactly those of (i) and (ii).